9. Write a C program for PT-109 American patrol boat, under the command of Lieutenant John F. Kennedy, was sunk by a Japanese destroyer, a message was received at an Australian wireless station in

Playfair code:

KXJEY UREBE ZWEHE WRYTU HEYFS

KREHE GOYFI WTTTU OLKSY CAJPO

BOTEI ZONTX BYBNT GONEY CUZWR

GDSON SXBOU YWRHE BAAHY USEDQ

Aim:

To decrypt a message encrypted using the Playfair cipher.

Procedure:

1. Read the keyword and ciphertext.
2. Construct the 5 × 5 Playfair matrix using the keyword.
3. Divide the ciphertext into pairs of letters.
4. Apply the Playfair decryption rules to each pair.
5. Display the decrypted plaintext.

Code:

```
#include <stdio.h>
#include <string.h>
#include <ctype.h>
int main() {
   char key[100], text[500], mat[26] = "";
   int used[26] = {0}, i, j = 0, k;
   printf("Enter keyword: ");
   scanf("%99s", key);
   printf("Enter ciphertext: ");
   scanf("%499s", text);
```

```
for (i = 0; key[i]; i++) {
    char c = toupper(key[i]);
    if (c == 'J') c = 'I';
    if (c >= 'A' && c <= 'Z' && !used[c-'A']) {
        mat[j++] = c;
        used[c-'A'] = 1;
    }
}
for (i = 0; i < 26; i++) {
    if (i == 9) continue;
    if (!used[i]) mat[j++] = 'A' + i;
}

for (i = 0; text[i]; i++)
    text[i] = toupper(text[i]) == 'J' ? 'I' :
            toupper(text[i]);
for (i = 0; text[i] && text[i+1]; i += 2) {
    int p = strchr(mat, text[i]) - mat;
    int q = strchr(mat, text[i+1]) - mat;
    int r1 = p / 5, c1 = p % 5;
    int r2 = q / 5, c2 = q % 5;
    if (r1 == r2) {
        text[i] = mat[r1*5 + (c1+4)%5];
        text[i+1] = mat[r2*5 + (c2+4)%5];
    } else if (c1 == c2) {
        text[i] = mat[((r1+4)%5)*5 + c1];
        text[i+1] = mat[((r2+4)%5)*5 + c2];
```

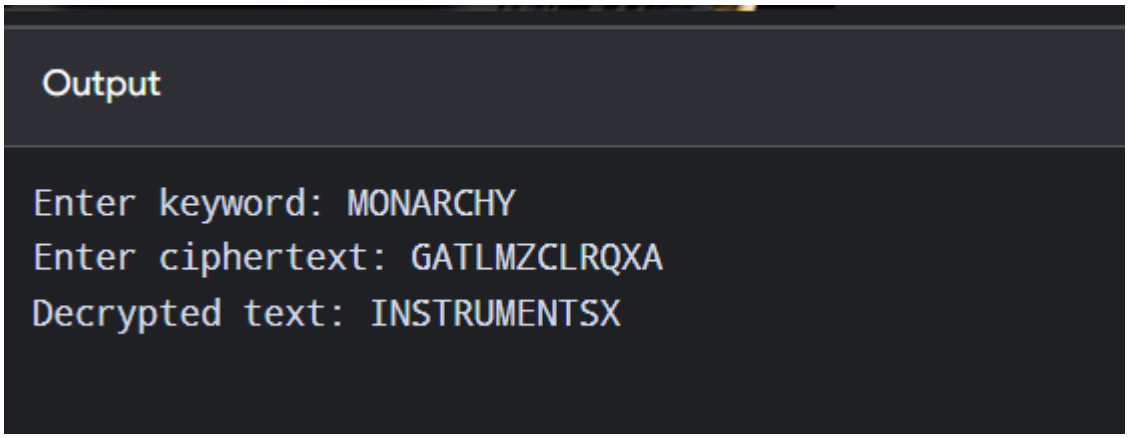
```
        } else {
            text[i] = mat[r1*5 + c2];
            text[i+1] = mat[r2*5 + c1];
        }
    }
    printf("Decrypted text: %s\n", text);
    return 0;
}
```

Input:

Enter keyword: MONARCHY

Enter ciphertext: GATLMZCLRQXA

Output:



Result:

Thus, the Playfair cipher decryption program was implemented successfully.